

# Fishery Independent Data Needs for US Caribbean Fisheries Stock Assessment

## Summary

A primary goal of the Southeast Fisheries Science Center's Caribbean Strategic Planning project is to reduce gaps in fishery independent datasets used to inform management. The first step in accomplishing that goal is to collect information on existing fishery independent surveys, and identify where major fishery-independent data gaps exist to improve fisheries management decisions in the US Caribbean. A working group was established to develop fishery-independent data inventories for three survey targets (reef fish, Caribbean spiny lobster, queen conch) with respect to methodology, spatial extent, and time series, as well as identify and describe time and space coverage gaps for three survey targets, and summarize findings. This document contains those findings.



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## Acronyms

| <i>Acronym</i> | <i>Definition</i>                                            |
|----------------|--------------------------------------------------------------|
| CIMAS          | Cooperative Institute for Marine and Atmospheric Studies     |
| CSP            | Caribbean Strategic Planning                                 |
| DCRMP          | Deep Coral Reef Monitoring Program                           |
| FI             | Fishery-independent                                          |
| NCRMP          | National Coral Reef Monitoring Program                       |
| NOAA           | National Oceanic and Atmospheric Administration              |
| NPS            | National Park Service                                        |
| PRCRMP         | Puerto Rico Coral Reef Monitoring Program                    |
| PR             | Puerto Rico                                                  |
| SEAMAP-C       | Southeast Area Monitoring and Assessment Program - Caribbean |
| SEFSC          | Southeast Fisheries Science Center                           |
| SERO           | Southeast Regional Office                                    |
| STTJ           | St. Thomas/St. John                                          |
| STX            | St. Croix                                                    |
| TCRMP          | Territorial Coral Reef Monitoring Program                    |
| UPR            | University of Puerto Rico                                    |
| USVI           | United States Virgin Islands                                 |
| UVI            | University of the Virgin Islands                             |

## Working Group

| <i>Participant</i> | <i>Email</i>                                                             | <i>Role</i> | <i>Affiliation</i> |
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Please contact the working group lead and coordinator with changes or suggestions.

## Introduction

Participants volunteered to join this strategic planning working group. During the initial meeting of the group, the participants developed their purpose, goals, and ideal outcomes, which are documented in the [Team Charter](#). The group met seventeen times over two years to accomplish the goals. Section A contains those goals and what was accomplished under each goal. Section B builds on the accomplished goals and contains specific data gaps identified in US Caribbean FI data.

## A. Goals & Summary of Actions to Address Goals:

1. Develop FI data inventories for the three survey targets (reef fish, Caribbean spiny lobster, queen conch) with respect to methodology, spatial extent, and time series.

Fishery-independent surveys employ standardized sampling methodology and techniques to assess changes in species abundance and other population metrics (size composition, age, etc.) through time and space. These data are generally used to assist with informing fisheries managers on the status of a given population. Based on the methodologies

employed, surveys can be classified as using extractive, non-extractive, or a combination of both methods to generate data. Extractive generally refers to removing animals from the water for biological samples to conduct age, growth, and reproductive studies. The life history information generated from these samples is often crucial to management. Non-extractive methods generally include counting (underwater or above), measuring (underwater or above), and/or weighing animals, resulting in the animals remaining alive and/or returning to the water alive. Oftentimes, surveys employ a combination of these methods to best generate data on species of interest. The type of survey methodology used (extractive vs. non-extractive) depends on the research needs and available resources. These data collected by fishery-independent surveys are used to calculate an index of abundance for target species and understand the size of the individuals in the population.

An inventory was developed to characterize current/ongoing FI data sources and the working group sought to collect information on the surveys regarding the target species group(s), methodology, and spatiotemporal range of the data collection. Programs with large-scale, long-term datasets were identified through working group member expert knowledge, input from colleagues, collaboration with other CSP working groups, and communication with relevant USVI and Puerto Rico staff. See Table 1 and the associated [spreadsheet](#) for additional information.

2. Identify and describe time/space coverage gaps for the three survey targets.

Summary tables were produced for each target species (Tables 2 – 4) to describe the survey sample frame, spatial extent, and data types related to each available and ongoing FI program. Separate meetings were held for each target species group to discuss the efficacy of each collection program for use in stock assessment, and to identify specific temporal and spatial coverage gaps (Table 5a – d and Section B). All recommendations were compiled for general data improvements (Section C).

3. Summarize findings in a written report.

This present report contains the findings of this group.

Table 1. Ongoing FI data collection programs in the US Virgin Islands and Puerto Rico.

| <i>Dataset</i>                                  | <i>Geographic Range</i> | <i>Data Points/Year</i> |
|-------------------------------------------------|-------------------------|-------------------------|
| SEAMAP-C Fish                                   | USVI & PR               | 2017 – ongoing          |
| SEAMAP-C Conch                                  | USVI & PR               | 2017 – ongoing          |
| SEAMAP-C Lobster                                | USVI & PR               | 2017 – ongoing          |
| Commercial Fishers Juvenile Lobster Trap Survey | PR                      | 2022 – ongoing          |
| Reef Fish Survey                                | PR                      | 2022 – ongoing          |
| Reef Fish Survey Expansion**                    | USVI                    | TBD                     |
| NPS Lobster Survey                              | USVI National Parks     | 2019 – ongoing          |
| NPS Seagrass/Conch Survey                       | USVI National Parks     | 2022 – ongoing          |
| NCRMP                                           | USVI & PR               | 2013 – ongoing          |
| TCRMP                                           | USVI                    | 2003 – ongoing          |
| PRCRMP                                          | PR                      | 1999 – ongoing          |
| Conch Surveys*                                  | USVI & PR               | 2023 – 2025             |
| DCRMP*                                          | USVI                    | 2020 – 2022             |

\*Short-term study

\*\*Upcoming survey

Table 2a. Reef fish FI surveys summary table.

| Reef Fish Summary |                          | SEAMAP-C                      | Deep Reef Fish Survey     | Shallow Reef Fish Survey | Reef Fish Survey Expansion** | NCRMP                     | TCRMP             | PRCRMP                       | DCRMP*                   |
|-------------------|--------------------------|-------------------------------|---------------------------|--------------------------|------------------------------|---------------------------|-------------------|------------------------------|--------------------------|
| Sample Frame      | Sample method            | Video, hook & line            | Video, hook & line        | Video, hook & line       | Hook & line                  | Diver                     | Diver             | Diver                        | Diver                    |
|                   | Survey type              | Combination                   | Combination               | Combination              | Combination                  | Non-Extractive            | Non-Extractive    | Non-Extractive               | Non-Extractive           |
|                   | Biosamples Taken         | Y                             | Y                         | Y                        | Y                            | N                         | N                 | N                            | N                        |
|                   | Primary Sample Unit      | 100x100 m                     | 500x500 m                 | 500x500 m                | 500x500 m                    | 50x50 m                   | 50x50 m           | 50x50 m                      | 50x50 m                  |
|                   | Secondary Sample Unit    | NA                            | NA                        | NA                       | NA                           | 15m diameter cylinder     | 25x4m transect    | 10x3m & 20x3m transect       | 15m diameter cylinder    |
|                   | Sampling design          | Random                        | Random                    | Random                   | Random                       | Random                    | Fixed             | Fixed                        | Random                   |
|                   | Spatial stratification   | Depth, habitat                | Depth, rugosity           | Depth, rugosity          | Depth, rugosity              | Depth, habitat, rugosity  | TCRMP specific    | Region, site, depth, habitat | Depth, habitat, rugosity |
|                   | Depth range              | 5 — 100 m                     | 50 — 450 m                | 20 — 50 m                | 20 — 450 m                   | 0 — 30 m                  | 5 — 65 m          | 2 — 35 m                     | 30 — 50 m                |
|                   | Geographic coverage      | W/SW & E PR, STTJ, STX        | PR island-wide            | PR island-wide           | STTJ, STX                    | PR, STTJ, STX island-wide | STTJ, STX         | PR                           | STTJ                     |
|                   | Sampling frequency       | 2 — 3 years in a 5-year cycle | Yearly                    | Yearly beginning 2025    | Yearly beginning TBD         | Every 2 years             | Yearly            | Every 2 years                | Yearly                   |
|                   | Planned sites            | 240                           | 200                       | 50                       | TBD                          | 250 STTJ, 250 STX, 300 PR | 34                | 42                           | 100                      |
|                   | Completed sites per year | 240                           | 200                       | 50                       | TBD                          | 250 STTJ, 250 STX, 250 PR | 34                | 42                           | 100                      |
| Data Type         | Relative abundance       | Y                             | Y                         | Y                        | Y                            | Y                         | Y                 | Y                            | Y                        |
|                   | Proportion occurrence    | Y                             | Y                         | Y                        | Y                            | Y                         | Y                 | Y                            | Y                        |
|                   | Length composition       | Y                             | Y                         | Y                        | Y                            | Y                         | Y                 | Y                            | Y                        |
|                   | Age composition          | Y                             | Y                         | Y                        | Y                            | N                         | N                 | N                            | N                        |
|                   | Sex                      | Y                             | Y                         | Y                        | Y                            | N                         | N                 | N                            | N                        |
|                   | Maturity                 | Limited                       | Limited                   | Limited                  | Limited                      | N                         | N                 | Limited                      | N                        |
|                   | Fecundity                | Limited                       | Limited                   | Limited                  | Limited                      | N                         | N                 | N                            | N                        |
|                   | Target phase             | Adults, limited juveniles     | Adults, limited juveniles | Adults, juveniles        | Adults, juveniles            | Adults, juveniles         | Adults, juveniles | Adults, juveniles            | Adults, juveniles        |
|                   | Benthic habitat          | Y                             | N                         | N                        | N                            | Y                         | Y                 | Y                            | Y                        |
| Contact           |                          | Walt Ingram, Eric Hoffmayer   | Kate Overly               | Kate Overly              | Kevin McCarthy               | Jay Grove                 | Tyler Smith       | Miguel Figuerola Hernandez   | Tyler Smith, Jay Grove   |

\*Short-term study

\*\*Upcoming survey

Table 2b. Puerto Rico reef fish FI surveys summary table.

| <b>Puerto Rico<br/>Reef Fish Summary</b> |                                 | <i>SEAMAP-C<br/>Est. 2017</i> | <i>Deep Reef Fish<br/>Survey<br/>Est. 2022</i> | <i>Shallow Reef Fish<br/>Survey<br/>Est. 2025</i> | <i>NCRMP<br/>Est. 2013</i> | <i>PRCRMP<br/>Est. 1999</i>          |
|------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------|---------------------------------------------------|----------------------------|--------------------------------------|
| <b>Sample Frame</b>                      | <i>Survey type</i>              | Video, hook and line          | Video, hook and line                           | Video, Hook and line                              | Diver                      | Diver                                |
|                                          | <i>Sample method</i>            | Combination                   | Combination                                    | Combination                                       | Non-Extractive             | Non-Extractive                       |
|                                          | <i>Biosamples taken</i>         | Y                             | Y                                              | Y                                                 | N                          | N                                    |
|                                          | <i>Primary Sample Unit</i>      | 100x100 m                     | 500x500 m                                      | 500x500 m                                         | 50x50 m                    | 50x50 m                              |
|                                          | <i>Secondary Sample Unit</i>    | NA                            | NA                                             | NA                                                | 15m diameter cylinder      | 10x3m & 20x3m transect               |
|                                          | <i>Sampling design</i>          | Random                        | Random                                         | Random                                            | Random                     | Fixed                                |
|                                          | <i>Spatial stratification</i>   | Depth, habitat                | Depth, rugosity                                | Depth, rugosity                                   | Depth, habitat, rugosity   | Region, site, depth, habitat         |
|                                          | <i>Depth range</i>              | 5 – 100 m                     | 50 – 450 m                                     | 20 – 50 m                                         | 0 – 30 m                   | 2 – 35 m                             |
|                                          | <i>Geographic coverage</i>      | W/SW & E                      | Island-wide                                    | Island-wide                                       | Island-wide                | Island-wide                          |
|                                          | <i>Sampling frequency</i>       | 2 – 3 years in a 5-year cycle | Yearly                                         | Yearly beginning 2025                             | Every 2 years              | Yearly in a 2-year cycle             |
|                                          | <i>Planned sites</i>            | 240                           | 200                                            | 50                                                | 300                        | 21                                   |
|                                          | <i>Completed sites per year</i> | 240                           | 200                                            | 50                                                | 250                        | 21 sites each year of 42 total sites |
| <b>Data Type</b>                         | <i>Relative abundance</i>       | Y                             | Y                                              | Y                                                 | Y                          | Y                                    |
|                                          | <i>Proportion occurrence</i>    | Y                             | Y                                              | Y                                                 | Y                          | Y                                    |
|                                          | <i>Length composition</i>       | Y                             | Y                                              | Y                                                 | Y                          | Y                                    |
|                                          | <i>Age composition</i>          | Y                             | Y                                              | Y                                                 | N                          | N                                    |
|                                          | <i>Sex</i>                      | Y                             | Y                                              | Y                                                 | N                          | N                                    |
|                                          | <i>Maturity</i>                 | Limited                       | Limited                                        | Limited                                           | N                          | Limited                              |
|                                          | <i>Fecundity</i>                | Limited                       | Limited                                        | Limited                                           | N                          | N                                    |
|                                          | <i>Target phase</i>             | Adults, limited juveniles     | Adults, juveniles                              | Adults, juveniles                                 | Adults, juveniles          | Adults, juveniles                    |
|                                          | <i>Benthic habitat</i>          | Y                             | N                                              | N                                                 | Y                          | Y                                    |
| <b>Contact</b>                           |                                 | Walt Ingram, Eric Hoffmayer   | Kate Overly                                    | Kate Overly                                       | Jay Grove                  | Miguel Figuerola Hernandez           |

Table 2c. St. Thomas/St. John reef fish FI surveys summary table

| <b>St. Thomas/St. John Reef Fish Summary</b> |                                 | <i>SEAMAP-C<br/>Est. 2017</i> | <i>Reef Fish Survey<br/>Expansion**<br/>TBD</i> | <i>NCRMP<br/>Est. 2013</i> | <i>TCRMP<br/>Est. 2003</i> | <i>DCRMP*<br/>2020 - 2022</i> |
|----------------------------------------------|---------------------------------|-------------------------------|-------------------------------------------------|----------------------------|----------------------------|-------------------------------|
| <b>Sample Frame</b>                          | <i>Survey type</i>              | Video, hook & line            | Hook & line                                     | Diver                      | Diver                      | Diver                         |
|                                              | <i>Sample method</i>            | Combination                   | Combination                                     | Non-Extractive             | Non-Extractive             | Non-Extractive                |
|                                              | <i>Biosamples taken</i>         | Y                             | Y                                               | N                          | N                          | N                             |
|                                              | <i>Primary Sample Unit</i>      | 100x100 m                     | 500x500 m                                       | 50x50 m                    | 50x50 m                    | 50x50 m                       |
|                                              | <i>Secondary Sample Unit</i>    | NA                            | NA                                              | 15m diameter cylinder      | 25x4m transect             | 15m diameter cylinder         |
|                                              | <i>Sampling design</i>          | Random                        | Random                                          | Random                     | Fixed                      | Random                        |
|                                              | <i>Spatial stratification</i>   | Depth, habitat                | Depth, rugosity                                 | Depth, habitat, rugosity   | TCRMP specific             | Depth, habitat, rugosity      |
|                                              | <i>Depth range</i>              | 5 – 100 m                     | 20 – 450 m                                      | 0 – 30 m                   | 5 – 65 m                   | 30 – 50 m                     |
|                                              | <i>Geographic coverage</i>      | Island wide                   | Island wide                                     | Island wide                | Island wide                | Island wide                   |
|                                              | <i>Sampling frequency</i>       | 2 – 3 years in a 5-year cycle | Yearly beginning TBD                            | Every 2 years              | Yearly                     | Yearly                        |
|                                              | <i>Planned sites</i>            | 120                           | TBD                                             | 250                        | 34                         | 100                           |
|                                              | <i>Completed sites per year</i> | 120                           | TBD                                             | 250                        | 34                         | 100                           |
| <b>Data Type</b>                             | <i>Relative abundance</i>       | Y                             | Y                                               | Y                          | Y                          | Y                             |
|                                              | <i>Proportion occurrence</i>    | Y                             | Y                                               | Y                          | Y                          | Y                             |
|                                              | <i>Length composition</i>       | Y                             | Y                                               | Y                          | Y                          | Y                             |
|                                              | <i>Age composition</i>          | Y                             | Y                                               | N                          | N                          | N                             |
|                                              | <i>Sex</i>                      | Y                             | Y                                               | N                          | N                          | N                             |
|                                              | <i>Maturity</i>                 | Limited                       | Limited                                         | N                          | N                          | N                             |
|                                              | <i>Fecundity</i>                | Limited                       | Limited                                         | N                          | N                          | N                             |
|                                              | <i>Target phase</i>             | Adults, limited juveniles     | Adults, juveniles                               | Adults, juveniles          | Adults, juveniles          | Adults, juveniles             |
|                                              | <i>Benthic habitat</i>          | Y                             | N                                               | Y                          | Y                          | Y                             |
| <b>Contact</b>                               |                                 | Walt Ingram, Eric Hoffmayer   | Kevin McCarthy                                  | Jay Grove                  | Tyler Smith                | Tyler Smith, Jay Grove        |

\*Short-term study

\*\*Upcoming survey

Table 2d. St. Croix reef fish FI surveys summary table

| <b>St. Croix Reef Fish Summary</b> |                                 | <i>SEAMAP-C<br/>Est. 2017</i> | <i>Reef Fish Survey Expansion**<br/>Est. 2026</i> | <i>NCRMP<br/>Est. 2013</i> | <i>TCRMP<br/>Est. 2003</i> |
|------------------------------------|---------------------------------|-------------------------------|---------------------------------------------------|----------------------------|----------------------------|
| <b>Sample Frame</b>                | <i>Survey type</i>              | Video, hook & line            | Hook & line                                       | Diver                      | Diver                      |
|                                    | <i>Sample method</i>            | Combination                   | Combination                                       | Non-Extractive             | Non-Extractive             |
|                                    | <i>Biosamples taken</i>         | Y                             | Y                                                 | N                          | N                          |
|                                    | <i>Primary Sample Unit</i>      | 100x100 m                     | 500x500 m                                         | 50x50 m                    | 50x50 m                    |
|                                    | <i>Secondary Sample Unit</i>    | NA                            | NA                                                | 15m diameter cylinder      | 25x4m transect             |
|                                    | <i>Sampling design</i>          | Random                        | Random                                            | Random                     | Fixed                      |
|                                    | <i>Spatial stratification</i>   | Depth, habitat                | Depth, rugosity                                   | Depth, habitat, rugosity   | TCRMP specific             |
|                                    | <i>Depth range</i>              | 5 — 100 m                     | 20 — 450 m                                        | 0 — 30 m                   | 5 — 65 m                   |
|                                    | <i>Geographic coverage</i>      | Island wide                   | Island wide                                       | Island wide                | Island wide                |
|                                    | <i>Sampling frequency</i>       | 2 — 3 years in a 5-year cycle | Yearly beginning TBD                              | Every 2 years              | Yearly                     |
|                                    | <i>Planned sites</i>            | 120                           | TBD                                               | 250                        | 34                         |
|                                    | <i>Completed sites per year</i> | 120                           | TBD                                               | 250                        | 34                         |
| <b>Data Type</b>                   | <i>Relative abundance</i>       | Y                             | Y                                                 | Y                          | Y                          |
|                                    | <i>Proportion occurrence</i>    | Y                             | Y                                                 | Y                          | Y                          |
|                                    | <i>Length composition</i>       | Y                             | Y                                                 | Y                          | Y                          |
|                                    | <i>Age composition</i>          | Y                             | Y                                                 | N                          | N                          |
|                                    | <i>Sex</i>                      | Y                             | Y                                                 | N                          | N                          |
|                                    | <i>Maturity</i>                 | Limited                       | Limited                                           | N                          | N                          |
|                                    | <i>Fecundity</i>                | Limited                       | Limited                                           | N                          | N                          |
|                                    | <i>Target phase</i>             | Adults, limited juveniles     | Adults, juveniles                                 | Adults, juveniles          | Adults, juveniles          |
|                                    | <i>Benthic habitat</i>          | Y                             | N                                                 | Y                          | Y                          |
| <b>Contact</b>                     |                                 | Walt Ingram, Eric Hoffmayer   | Kevin McCarthy                                    | Jay Grove                  | Tyler Smith                |

\*\*Upcoming survey

Table 3a. Spiny lobster FI surveys summary table

| <b>Spiny Lobster Summary</b> |                               | <i>NPS Lobster Survey<br/>Est. 2019</i>   | <i>SEAMAP-C (USVI)<br/>Est. 2017</i> | <i>SEAMAP-C (PR)<br/>Est. 2017</i>                | <i>Commercial Fishers<br/>Juvenile Lobster Trap<br/>Survey<br/>Est. 2022</i> |
|------------------------------|-------------------------------|-------------------------------------------|--------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------|
| <b>Sample frame</b>          | <i>Survey type</i>            | Diver                                     | Diver                                | Diver                                             | Lobster trap                                                                 |
|                              | <i>Sample method</i>          | Non-Extractive                            | Combination                          | Combination                                       | Non-Extractive                                                               |
|                              | <i>Biosamples Taken</i>       | N                                         | Y                                    | Y                                                 | N                                                                            |
|                              | <i>Primary Sample Unit</i>    | 353 m <sup>2</sup>                        | Variable                             | NA                                                | 50x50 m                                                                      |
|                              | <i>Secondary Sample Unit</i>  | NA                                        | NA                                   | NA                                                | NA                                                                           |
|                              | <i>Sampling design</i>        | Random                                    | Random                               | Haphazard;<br>Non-standardized<br>random swimming | Random                                                                       |
|                              | <i>Spatial stratification</i> | Depth, habitat                            | Habitat                              | Depth                                             | Depth, habitat                                                               |
|                              | <i>Depth range</i>            | 0 – 30 m                                  | 0 – 30 m                             | 0 – 27 m                                          | 0 – 40 m                                                                     |
|                              | <i>Geographic coverage</i>    | National parks in USVI                    | STTJ, STX                            | W/SW PR                                           | PR island-wide                                                               |
|                              | <i>Sampling frequency</i>     | 1 year in a 4-year cycle<br>for each park | 2 years in a 5-year<br>cycle         | 1 year in a 5-year cycle                          | Yearly                                                                       |
|                              | <i>Planned sites</i>          | 100                                       | 80 STTJ<br>80 in STX                 | -                                                 | TBD                                                                          |
|                              | <i>Completed sites</i>        | 85 – 100                                  | 80 STTJ<br>80 in STX                 | -                                                 | TBD                                                                          |
| <b>Data type</b>             | <i>Relative abundance</i>     | Y                                         | Y                                    | N                                                 | Y                                                                            |
|                              | <i>Proportion occurrence</i>  | Y                                         | Y                                    | N                                                 | Y                                                                            |
|                              | <i>Length composition</i>     | Y                                         | Y                                    | Y                                                 | Y                                                                            |
|                              | <i>Age composition</i>        | N                                         | Y                                    | Y                                                 | N                                                                            |
|                              | <i>Sex</i>                    | Y                                         | Y                                    | Y                                                 | Y                                                                            |
|                              | <i>Maturity</i>               | Y                                         | Limited                              | N                                                 | Y                                                                            |
|                              | <i>Fecundity</i>              | N                                         | Limited                              | N                                                 | N                                                                            |
|                              | <i>Target phase</i>           | Adults, limited juveniles                 | Adults, juveniles                    | Adults, juveniles                                 | Juveniles                                                                    |
|                              | <i>Benthic habitat</i>        | Y                                         | Limited                              | Y                                                 | Limited                                                                      |
| <b>Contact</b>               |                               | Mike Feeley                               | JJ Cruz-Mota                         | JJ Cruz-Mota                                      | Steve Smith, Kevin McCarthy                                                  |

Table 3b. Puerto Rico spiny lobster FI surveys summary table

| <b><i>Puerto Rico Spiny Lobster Summary</i></b> |                               | <i>SEAMAP-C (PR)<br/>Est. 2017</i>                    | <i>Commercial Fishers Juvenile Lobster Trap<br/>Survey<br/>Est. 2022</i> |
|-------------------------------------------------|-------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
| <b><i>Sample frame</i></b>                      | <i>Survey type</i>            | Diver                                                 | Lobster trap                                                             |
|                                                 | <i>Sample method</i>          | Combination                                           | Non-Extractive                                                           |
|                                                 | <i>Biosamples Taken</i>       | Y                                                     | N                                                                        |
|                                                 | <i>Primary Sample Unit</i>    | NA                                                    | 50x50 m                                                                  |
|                                                 | <i>Secondary Sample Unit</i>  | NA                                                    | NA                                                                       |
|                                                 | <i>Sampling design</i>        | Haphazard selection; Non-standardized random swimming | Random                                                                   |
|                                                 | <i>Spatial stratification</i> | Depth                                                 | Depth, habitat                                                           |
|                                                 | <i>Depth range</i>            | 0 – 27 m                                              | 0 – 40 m                                                                 |
|                                                 | <i>Geographic coverage</i>    | W/SW                                                  | Island wide                                                              |
|                                                 | <i>Sampling frequency</i>     | 1 year in a 5-year cycle                              | Yearly                                                                   |
|                                                 | <i>Planned sites</i>          | -                                                     | TBD                                                                      |
|                                                 | <i>Completed sites</i>        | -                                                     | TBD                                                                      |
| <b><i>Data types</i></b>                        | <i>Relative abundance</i>     | N                                                     | Y                                                                        |
|                                                 | <i>Proportion occurrence</i>  | N                                                     | Y                                                                        |
|                                                 | <i>Length composition</i>     | Y                                                     | Y                                                                        |
|                                                 | <i>Age composition</i>        | Y                                                     | N                                                                        |
|                                                 | <i>Sex</i>                    | Y                                                     | Y                                                                        |
|                                                 | <i>Maturity</i>               | N                                                     | Y                                                                        |
|                                                 | <i>Fecundity</i>              | N                                                     | N                                                                        |
|                                                 | <i>Target phase</i>           | Adults, juveniles                                     | Juveniles                                                                |
|                                                 | <i>Benthic habitat</i>        | Y                                                     | Limited                                                                  |
| <b><i>Contact</i></b>                           |                               | JJ Cruz-Mota                                          | Steve Smith, Kevin McCarthy                                              |

Table 3c. St. Thomas/St. John spiny lobster FI surveys summary table

| <b>St. Thomas/St. John Spiny Lobster Summary</b> |                               | <i>NPS Lobster Survey<br/>Est. 2019</i> | <i>SEAMAP-C (USVI)<br/>Est. 2017</i> |
|--------------------------------------------------|-------------------------------|-----------------------------------------|--------------------------------------|
| <i>Sample Frame</i>                              | <i>Survey type</i>            | Diver                                   | Diver                                |
|                                                  | <i>Sample method</i>          | Non-Extractive                          | Combination                          |
|                                                  | <i>Biosamples Taken</i>       | N                                       | Y                                    |
|                                                  | <i>Primary Sample Unit</i>    | 353 m <sup>2</sup>                      | Variable                             |
|                                                  | <i>Secondary Sample Unit</i>  | NA                                      | NA                                   |
|                                                  | <i>Sampling design</i>        | Random                                  | Random                               |
|                                                  | <i>Spatial stratification</i> | Depth, habitat                          | Habitat                              |
|                                                  | <i>Depth range</i>            | 0 – 30 m                                | 0 – 30 m                             |
|                                                  | <i>Geographic coverage</i>    | Virgin Islands National Park            | Island-wide                          |
|                                                  | <i>Sampling frequency</i>     | 1 year in a 4-year cycle for each park  | 1 year in a 5-year cycle             |
|                                                  | <i>Planned sites</i>          | 100                                     | 80                                   |
|                                                  | <i>Completed sites</i>        | 85 – 100                                | 80                                   |
| <i>Data Type</i>                                 | <i>Relative abundance</i>     | Y                                       | Y                                    |
|                                                  | <i>Proportion occurrence</i>  | Y                                       | Y                                    |
|                                                  | <i>Length composition</i>     | Y                                       | Y                                    |
|                                                  | <i>Age composition</i>        | N                                       | Y                                    |
|                                                  | <i>Sex</i>                    | Y                                       | Y                                    |
|                                                  | <i>Maturity</i>               | Y                                       | Limited                              |
|                                                  | <i>Fecundity</i>              | N                                       | Limited                              |
|                                                  | <i>Target phase</i>           | Adults, limited juveniles               | Adults, juveniles                    |
|                                                  | <i>Benthic habitat</i>        | Y                                       | Limited                              |
| <i>Contact</i>                                   |                               | Mike Feeley                             | JJ Cruz-Mota                         |

Table 3d. St. Croix spiny lobster FI surveys summary table

| <b>St. Croix Spiny Lobster Summary</b> |                               | <i>NPS Lobster Survey<br/>Est. 2019</i>                                                  | <i>SEAMAP-C (USVI)<br/>Est. 2017</i> |
|----------------------------------------|-------------------------------|------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Sample Frame</b>                    | <i>Survey type</i>            | Diver                                                                                    | Diver                                |
|                                        | <i>Sample method</i>          | Non-Extractive                                                                           | Combination                          |
|                                        | <i>Biosamples Taken</i>       | N                                                                                        | Y                                    |
|                                        | <i>Primary Sample Unit</i>    | 353 m <sup>2</sup>                                                                       | Variable                             |
|                                        | <i>Secondary Sample Unit</i>  | NA                                                                                       | NA                                   |
|                                        | <i>Sampling design</i>        | Random                                                                                   | Random                               |
|                                        | <i>Spatial stratification</i> | Depth, habitat                                                                           | Habitat                              |
|                                        | <i>Depth range</i>            | 0 – 30 m                                                                                 | 0 – 30 m                             |
|                                        | <i>Geographic coverage</i>    | Buck Island Reef National Monument/Salt River Bay Historical Park and Ecological Reserve | Island wide                          |
|                                        | <i>Sampling frequency</i>     | 1 year in a 4-year cycle for each park                                                   | 1 year in a 5-year cycle             |
|                                        | <i>Planned sites</i>          | 100                                                                                      | 80                                   |
|                                        | <i>Completed sites</i>        | 85 – 100                                                                                 | 80                                   |
| <b>Data Type</b>                       | <i>Relative abundance</i>     | Y                                                                                        | Y                                    |
|                                        | <i>Proportion occurrence</i>  | Y                                                                                        | Y                                    |
|                                        | <i>Length composition</i>     | Y                                                                                        | Y                                    |
|                                        | <i>Age composition</i>        | N                                                                                        | Y                                    |
|                                        | <i>Sex</i>                    | Y                                                                                        | Y                                    |
|                                        | <i>Maturity</i>               | Y                                                                                        | Limited                              |
|                                        | <i>Fecundity</i>              | N                                                                                        | Limited                              |
|                                        | <i>Target phase</i>           | Adults, limited juveniles                                                                | Adults, juveniles                    |
|                                        | <i>Benthic habitat</i>        | Y                                                                                        | Limited                              |
| <b>Contact</b>                         |                               | Mike Feeley                                                                              | JJ Cruz-Mota                         |

Table 4a. Queen conch FI surveys summary table

| <i>Queen Conch Summary</i> |                        | <i>SEAMAP-C<br/>Est. 2017</i>                     | <i>NPS Lobster<br/>Survey***<br/>Est. 2019</i> | <i>NPS Seagrass<br/>Survey***<br/>Est. 2022</i> | <i>Conch Surveys*<br/>2023 – 2025</i> |
|----------------------------|------------------------|---------------------------------------------------|------------------------------------------------|-------------------------------------------------|---------------------------------------|
| Sample Frame               | Survey type            | Diver                                             | Diver                                          | Diver                                           | Diver                                 |
|                            | Sample method          | Non-extractive; Belt transect to radial survey    | Non-extractive                                 | Non-extractive                                  | Non-extractive; Radial survey         |
|                            | Biosamples Taken       | N                                                 | N                                              | N                                               | N                                     |
|                            | Primary Sample Unit    | NA                                                | 2500 m <sup>2</sup>                            | NA                                              | NA                                    |
|                            | Secondary Sample Unit  | NA                                                | 353 m <sup>2</sup>                             | 177 m <sup>2</sup>                              | 314 m <sup>2</sup>                    |
|                            | Sampling design        | Random                                            | Random                                         | Random                                          | Random                                |
|                            | Spatial stratification | Depth, habitat                                    | Depth, habitat                                 | Depth                                           | Depth, habitat                        |
|                            | Depth range            | 0 – 30 m                                          | 0 – 30 m                                       | 0 – 25 m                                        | 0 – 30 m                              |
|                            | Geographic coverage    | E & W PR, USVI                                    | National parks in USVI                         | National parks in USVI                          | STTJ, STX, PR                         |
|                            | Sampling frequency     | 1 year in a 5-year cycle (PR), infrequent in USVI | One park per year in a 4-year cycle            | One park per year in a 3-year cycle             | One island per year                   |
|                            | Planned sites          | -                                                 | 100                                            | 100                                             | -                                     |
|                            | Completed sites        | -                                                 | 85 – 100                                       | 100                                             | -                                     |
| Data Type                  | Relative abundance     | Y                                                 | Y                                              | Y                                               | Y                                     |
|                            | Proportion occurrence  | Y                                                 | Y                                              | Y                                               | Y                                     |
|                            | Length composition     | Limited                                           | N                                              | N                                               | Y                                     |
|                            | Age composition        | Limited                                           | N                                              | N                                               | Limited                               |
|                            | Sex                    | N                                                 | N                                              | N                                               | N                                     |
|                            | Maturity               | N                                                 | Y                                              | Y                                               | Limited                               |
|                            | Fecundity              | N                                                 | N                                              | N                                               | N                                     |
|                            | Target phase           | Adults, juveniles                                 | Adults, juveniles                              | Adults, juveniles                               | Adults, juveniles                     |
|                            | Benthic habitat        | Y                                                 | Y                                              | Y                                               | Y                                     |
| Contact                    |                        | JJ Cruz-Mota                                      | Mike Feeley                                    | Mike Feeley                                     | Jennifer Doerr                        |

\*Short-term study

\*\*\*Conch as incidental

Table 4b. Puerto Rico queen conch FI surveys summary table

| <b>Puerto Rico Queen Conch Summary</b> |                               | <b>SEAMAP-C (PR)<br/>Est. 2017</b>             | <b>Conch Surveys*<br/>2023 – 2025</b> |
|----------------------------------------|-------------------------------|------------------------------------------------|---------------------------------------|
| <b>Sample Frame</b>                    | <i>Survey type</i>            | Diver                                          | Diver                                 |
|                                        | <i>Sample method</i>          | Non-extractive; Belt transect to radial survey | Non-extractive; Radial survey         |
|                                        | <i>Biosamples Taken</i>       | N                                              | N                                     |
|                                        | <i>Primary Sample Unit</i>    | Circular transect of 10 m radius               | NA                                    |
|                                        | <i>Secondary Sample Unit</i>  | NA                                             | 314 m <sup>2</sup>                    |
|                                        | <i>Sampling design</i>        | Random                                         | Random                                |
|                                        | <i>Spatial stratification</i> | Depth, habitat                                 | Depth, habitat                        |
|                                        | <i>Depth range</i>            | 0 – 30 m                                       | 0 – 30 m                              |
|                                        | <i>Geographic coverage</i>    | E & W                                          | Island wide                           |
|                                        | <i>Sampling frequency</i>     | 1 year in a 5-year cycle                       | 1 year in a 3-year cycle              |
|                                        | <i>Planned sites</i>          | -                                              | -                                     |
|                                        | <i>Completed sites</i>        | -                                              | -                                     |
| <b>Data Type</b>                       | <i>Relative abundance</i>     | Y                                              | Y                                     |
|                                        | <i>Proportion occurrence</i>  | Y                                              | Y                                     |
|                                        | <i>Length composition</i>     | Limited                                        | Y                                     |
|                                        | <i>Age composition</i>        | Limited                                        | Limited                               |
|                                        | <i>Sex</i>                    | N                                              | N                                     |
|                                        | <i>Maturity</i>               | N                                              | Limited                               |
|                                        | <i>Fecundity</i>              | N                                              | N                                     |
|                                        | <i>Target phase</i>           | Adults, juveniles                              | Adults, juveniles                     |
|                                        | <i>Benthic habitat</i>        | Y                                              | Y                                     |
| <b>Contact</b>                         |                               | JJ Cruz-Mota                                   | Jennifer Doerr                        |

\*Short-term study

Table 4c. St. Thomas/St. John queen conch FI surveys summary table

| <b>St. Thomas/St. John Queen Conch Summary</b> |                               | <b>SEAMAP-C (USVI)<br/>Est. 2017</b>           | <b>NPS Lobster Survey***<br/>Est. 2019</b> | <b>NPS Seagrass Survey***<br/>Est. 2022</b> | <b>Conch Surveys*<br/>2023 – 2025</b> |
|------------------------------------------------|-------------------------------|------------------------------------------------|--------------------------------------------|---------------------------------------------|---------------------------------------|
| <b>Sample Frame</b>                            | <i>Survey type</i>            | Diver                                          | Diver                                      | Diver                                       | Diver                                 |
|                                                | <i>Sample method</i>          | Non-extractive; Belt transect to radial survey | Non-extractive                             | Non-extractive                              | Non-extractive; radial survey         |
|                                                | <i>Biosamples Taken</i>       | N                                              | N                                          | N                                           | N                                     |
|                                                | <i>Primary Sample Unit</i>    | NA                                             | 2500 m <sup>2</sup>                        | NA                                          | NA                                    |
|                                                | <i>Secondary Sample Unit</i>  | NA                                             | 353 m <sup>2</sup>                         | 177 m <sup>2</sup>                          | 314 m <sup>2</sup>                    |
|                                                | <i>Sampling design</i>        | Random                                         | Random                                     | Random                                      | Random                                |
|                                                | <i>Spatial stratification</i> | Depth, habitat                                 | Depth, habitat                             | Depth                                       | Depth, habitat                        |
|                                                | <i>Depth range</i>            | 0 – 30 m                                       | 0 – 30 m                                   | 0 – 25 m                                    | 0 – 30 m                              |
|                                                | <i>Geographic coverage</i>    | Island-wide                                    | Virgin Islands National Park               | Virgin Islands National Park                | Island-wide                           |
|                                                | <i>Sampling frequency</i>     | 1 year in a 5-year cycle                       | 1 year in a 4-year cycle                   | 1 year in a 3-year cycle                    | 1 year in a 3 year cycle              |
|                                                | <i>Planned sites</i>          | -                                              | 100                                        | 100                                         | -                                     |
|                                                | <i>Completed sites</i>        | -                                              | 85–100                                     | 100                                         | -                                     |
| <b>Data Type</b>                               | <i>Relative abundance</i>     | Y                                              | Y                                          | Y                                           | Y                                     |
|                                                | <i>Proportion occurrence</i>  | Y                                              | Y                                          | Y                                           | Y                                     |
|                                                | <i>Length composition</i>     | Limited                                        | N                                          | N                                           | Y                                     |
|                                                | <i>Age composition</i>        | Limited                                        | N                                          | N                                           | Limited                               |
|                                                | <i>Sex</i>                    | N                                              | N                                          | N                                           | N                                     |
|                                                | <i>Maturity</i>               | N                                              | Y                                          | Y                                           | Limited                               |
|                                                | <i>Fecundity</i>              | N                                              | N                                          | N                                           | N                                     |
|                                                | <i>Target phase</i>           | Adults, juveniles                              | Adults, juveniles                          | Adults, juveniles                           | Adults, juveniles                     |
|                                                | <i>Benthic habitat</i>        | Y                                              | Y                                          | Y                                           | Y                                     |
| <b>Contact</b>                                 |                               | JJ Cruz-Mota                                   | Mike Feeley                                | Mike Feeley                                 | Jennifer Doerr                        |

\*Short-term study

\*\*\*Conch as incidental

Table 4d. St. Croix queen conch FI surveys summary table

| <b>St. Croix Queen Conch Summary</b> |                               | <b>SEAMAP-C<br/>Est. 2017</b> | <b>NPS Lobster<br/>Survey***<br/>Est. 2019</b>                                           | <b>NPS Seagrass<br/>Survey***<br/>Est. 2022</b>                                          | <b>Conch Surveys*<br/>2023 – 2025</b> |
|--------------------------------------|-------------------------------|-------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Sample Frame</b>                  | <i>Survey type</i>            | Diver                         | Diver                                                                                    | Diver                                                                                    | Diver                                 |
|                                      | <i>Sample method</i>          | Non-Extractive                | Non-Extractive                                                                           | Non-Extractive                                                                           | Non-Extractive; radial survey         |
|                                      | <i>Biosamples Taken</i>       | N                             | N                                                                                        | N                                                                                        | N                                     |
|                                      | <i>Primary Sample Unit</i>    | NA                            | 2500 m <sup>2</sup>                                                                      | NA                                                                                       | NA                                    |
|                                      | <i>Secondary Sample Unit</i>  | NA                            | 353 m <sup>2</sup>                                                                       | 177 m <sup>2</sup>                                                                       | 314 m <sup>2</sup>                    |
|                                      | <i>Sampling design</i>        | Random                        | Random                                                                                   | Random                                                                                   | Random                                |
|                                      | <i>Spatial stratification</i> | Depth, habitat                | Depth, habitat                                                                           | Depth                                                                                    | Depth, habitat                        |
|                                      | <i>Depth range</i>            | 0 – 30 m                      | 0 – 30 m                                                                                 | 0 – 25 m                                                                                 | 0 – 30 m                              |
|                                      | <i>Geographic coverage</i>    | Island wide                   | Buck Island Reef National Monument/Salt River Bay Historical Park and Ecological Reserve | Buck Island Reef National Monument/Salt River Bay Historical Park and Ecological Reserve | Island-wide                           |
|                                      | <i>Sampling frequency</i>     | 1 year in a 5-year cycle      | 1 year in a 4-year cycle                                                                 | 1 year in a 3-year cycle                                                                 | 1 year in a 3 year cycle              |
|                                      | <i>Planned sites</i>          | -                             | 100                                                                                      | 100                                                                                      | -                                     |
|                                      | <i>Completed sites</i>        | -                             | 85 – 100                                                                                 | 100                                                                                      | -                                     |
| <b>Data Type</b>                     | <i>Relative abundance</i>     | Y                             | Y                                                                                        | Y                                                                                        | Y                                     |
|                                      | <i>Proportion occurrence</i>  | Y                             | Y                                                                                        | Y                                                                                        | Y                                     |
|                                      | <i>Length composition</i>     | Limited                       | N                                                                                        | N                                                                                        | Y                                     |
|                                      | <i>Age composition</i>        | Limited                       | N                                                                                        | N                                                                                        | Limited                               |
|                                      | <i>Sex</i>                    | N                             | N                                                                                        | N                                                                                        | N                                     |
|                                      | <i>Maturity</i>               | N                             | Y                                                                                        | Y                                                                                        | Limited                               |
|                                      | <i>Fecundity</i>              | N                             | N                                                                                        | N                                                                                        | N                                     |
|                                      | <i>Target phase</i>           | Adults, juveniles             | Adults, juveniles                                                                        | Adults, juveniles                                                                        | Adults, juveniles                     |
|                                      | <i>Benthic habitat</i>        | Y                             | Y                                                                                        | Y                                                                                        | Y                                     |
| <b>Contact</b>                       |                               | JJ Cruz-Mota                  | Mike Feeley                                                                              | Mike Feeley                                                                              | Jennifer Doerr                        |

\*Short-term study

\*\*\*Conch as incidental

Table 5a. Coverage gaps for FI survey targets in the US Virgin Islands and Puerto Rico. An 'X' indicates there is no gap in data coverage.

| <b>Coverage Gaps</b> |                              | <i>Reef Fish</i>                              | <i>Spiny Lobster</i>                    | <i>Queen Conch</i>                                                                               |
|----------------------|------------------------------|-----------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| <i>Sample Frame</i>  | <i>Geographic coverage</i>   | X                                             | X                                       | X<br>(High density/low density as well as stratified random sampling but need continued support) |
|                      | <i>Depth coverage</i>        | X                                             | No surveys beyond 40 m                  | X                                                                                                |
|                      | <i>Time coverage</i>         | Seasonality missing or not adequately sampled | Unclear if survey frequency is adequate | Not adequately sampled                                                                           |
| <i>Data Type</i>     | <i>Relative abundance</i>    | X                                             | X                                       | X                                                                                                |
|                      | <i>Proportion occurrence</i> | X                                             | X                                       | X                                                                                                |
|                      | <i>Length composition</i>    | X                                             | X                                       | X                                                                                                |
|                      | <i>Age composition</i>       | X                                             | X                                       | Cannot currently age                                                                             |
|                      | <i>Sex</i>                   | X                                             | X                                       | Missing or not adequately sampled                                                                |
|                      | <i>Maturity</i>              | Missing or not adequately sampled             | Missing or not adequately sampled       | Missing or not adequately sampled                                                                |
|                      | <i>Fecundity</i>             | Missing or not adequately sampled             | Missing or not adequately sampled       | Missing or not adequately sampled                                                                |
|                      | <i>Juveniles sampled</i>     | Missing or not adequately sampled             | Missing or not adequately sampled       | 0 - 1 year old missing or not adequately sampled; 5 - 6 cm sampled                               |
|                      | <i>Benthic habitat</i>       | Missing or not adequately sampled             | X                                       | X                                                                                                |

Table 5b. Coverage gaps for Reef Fish across Island Platforms. An 'X' indicates there is no gap in data coverage.

| Coverage Gaps for Reef Fish |                       | Puerto Rico                                    |             | STTJ                                               |            | STX                                                |            |
|-----------------------------|-----------------------|------------------------------------------------|-------------|----------------------------------------------------|------------|----------------------------------------------------|------------|
|                             |                       | Note                                           | Coverage =X | Note                                               | Coverage=X | Note                                               | Coverage=X |
| Sampling frame              | Geographic coverage   | Island-wide                                    | X           | Island-wide                                        | X          | Island-wide                                        | X          |
|                             | Depth coverage        | 0 – 450 m                                      | X           | 0 – 100 m                                          | –          | 0 – 100 m                                          | –          |
|                             | Time coverage         | 20 – 450 m yearly ; < 20 m ~every 2 years      | –           | 0 – 100 m ~every 3 – 5 years; 0 – 450 m yearly TBD | –          | 0 – 100 m ~every 3 – 5 years; 0 – 450 m yearly TBD | –          |
| Data types                  | Relative abundance    | Data collected via visual & extractive surveys | X           | Data collected via visual & extractive surveys     | X          | Data collected via visual & extractive surveys     | X          |
|                             | Proportion occurrence | Data collected via visual & extractive surveys | X           | Data collected via visual & extractive surveys     | X          | Data collected via visual & extractive surveys     | X          |
|                             | Length composition    | Data collected via visual & extractive surveys | X           | Data collected via visual & extractive surveys     | X          | Data collected via visual & extractive surveys     | X          |
|                             | Age composition       | Data collected via extractive surveys          | X           | Data collected via extractive surveys              | X          | Data collected via extractive surveys              | X          |
|                             | Sex                   | Data collected via extractive surveys          | X           | Data collected via extractive surveys              | X          | Data collected via extractive surveys              | X          |
|                             | Maturity              | Limited data                                   | –           | Limited data                                       | –          | Limited data                                       | –          |
|                             | Fecundity             | Limited data                                   | –           | Limited data                                       | –          | Limited data                                       | –          |
|                             | Target phase          | Data collected on all phases                   | X           | Data collected on all phases                       | X          | Data collected on all phases                       | X          |
|                             | Benthic habitat       | Data collected via visual surveys              | X           | Data collected via visual surveys                  | X          | Data collected via visual surveys                  | X          |

Table 5c. Coverage gaps for Spiny Lobster across Island Platforms. An 'X' indicates there is no gap in data coverage.

| Coverage Gaps for Spiny Lobster |                       | Puerto Rico                                    |            | STTJ                                                                              |            | STX                                                                               |            |
|---------------------------------|-----------------------|------------------------------------------------|------------|-----------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------|------------|
|                                 |                       | Note                                           | Coverage=X | Note                                                                              | Coverage=X | Note                                                                              | Coverage=X |
| Sample Frame                    | Geographic coverage   | Island-wide by one survey                      | X          | Island-wide by one survey                                                         | –          | Island-wide by one survey                                                         | –          |
|                                 | Depth coverage        | 0 – 40 m                                       | –          | 0 – 30 m                                                                          | –          | 0 – 30 m                                                                          | –          |
|                                 | Time coverage         | 0 – 27 m ~ every 5 years; 0 – 40 m yearly      | –          | 0 – 30 m 2 years in a 5-year cycle island-wide; 0 – 30 m yearly in national parks | –          | 0 – 30 m 2 years in a 5-year cycle island-wide; 0 – 30 m yearly in national parks | –          |
| Data Type                       | Relative abundance    | Data collected by one survey                   | –          | Data collected via visual & extractive surveys                                    | X          | Data collected via visual & extractive surveys                                    | X          |
|                                 | Proportion occurrence | Data collected by one survey                   | –          | Data collected via visual & extractive surveys                                    | X          | Data collected via visual & extractive surveys                                    | X          |
|                                 | Length composition    | Data collected via visual & extractive surveys | X          | Data collected via visual & extractive surveys                                    | X          | Data collected via visual & extractive surveys                                    | X          |
|                                 | Age composition       | Data collected via visual survey every 5 years | –          | Data collected via extractive survey                                              | X          | Data collected via extractive survey                                              | X          |
|                                 | Sex                   | Data collected via visual & extractive surveys | X          | Data collected via visual & extractive surveys                                    | X          | Data collected via visual & extractive surveys                                    | X          |
|                                 | Maturity              | Data collected via visual survey every 5 years | –          | Limited data                                                                      | –          | Limited data                                                                      | –          |
|                                 | Fecundity             | Data not collected                             | –          | Limited data                                                                      | –          | Limited data                                                                      | –          |
|                                 | Target phase          | Data on adults collected every 5 years         | –          | Limited data                                                                      | –          | Limited data                                                                      | –          |
|                                 | Benthic habitat       | Limited data                                   | –          | Limited data                                                                      | –          | Limited data                                                                      | –          |

Table 5d. Coverage gaps for Queen Conch across Island Platforms. An 'X' indicates there is no gap in data coverage.

| Coverage Gaps for Queen Conch |                       | Puerto Rico                       |            | STTJ                                                     |            | STX                                                      |            |
|-------------------------------|-----------------------|-----------------------------------|------------|----------------------------------------------------------|------------|----------------------------------------------------------|------------|
|                               |                       | Note                              | Coverage=X | Note                                                     | Coverage=X | Note                                                     | Coverage=X |
| Sample Frame                  | Geographic coverage   | Island-wide by one survey         | –          | Island-wide                                              | X          | Island-wide                                              | X          |
|                               | Depth coverage        | 0 – 30 m                          |            | 0 – 30 m                                                 | X          | 0 – 30 m                                                 | X          |
|                               | Time coverage         | 0 – 30 m ~ every 3 & 5 years      |            | 0 – 30 m ~ every 3, 4, & 5 years; 0 – 25 m every 4 years | X          | 0 – 30 m ~ every 3, 4, & 5 years; 0 – 25 m every 4 years | X          |
| Data Type                     | Relative abundance    | Data collected via visual surveys | X          | Data collected via visual surveys                        | X          | Data collected via visual surveys                        | X          |
|                               | Proportion occurrence | Data collected via visual surveys | X          | Data collected via visual surveys                        | X          | Data collected via visual surveys                        | X          |
|                               | Length composition    | Limited data                      | –          | Limited data                                             | –          | Limited data                                             | –          |
|                               | Age composition       | Limited data                      | –          | Limited data                                             | –          | Limited data                                             | –          |
|                               | Sex                   | No data collected                 | –          | No data collected                                        | –          | No data collected                                        | –          |
|                               | Maturity              | Limited data                      | –          | Limited data                                             | –          | Limited data                                             | –          |
|                               | Fecundity             | No data collected                 | –          | No data collected                                        | –          | No data collected                                        | –          |
|                               | Target phase          | Data collected on all phases      | X          | Data collected on all phases                             | X          | Data collected on all phases                             | X          |
|                               | Benthic habitat       | Data collected via visual surveys | X          | Data collected via visual surveys                        | X          | Data collected via visual surveys                        | X          |

## B. Summary of Fishery Independent Data Gaps:

### 1. Reef Fish

#### a. Puerto Rico

When all five reef fish sampling programs in Puerto Rico are considered, sampling has occurred at least every two years, island-wide since 2013 with at least one survey type (Table 2b). Other programs which sample fewer sites every two to three years have been in operation since 1999 and 2013 (Table 2b). Starting in 2025, one program began sampling yearly, island-wide and that project is ongoing (Table 2b). As such, the planned geographic coverage for shallower depths (0 – 100 m) appears to be adequate, pending funding. Another program focused on deeper reef fish habitat (50 – 450 m) has sampled island-wide annually since 2022 (Table 2b).

The sampling frequency of the survey types may be missing or inadequate. Specifically, information regarding the seasonality of species cannot always be obtained due to the sampling schedules. Among the surveys, the following data types are sampled annually, island-wide, with at least one survey type from 20 – 450 m: relative abundance, proportion occurrence, length composition, age composition, and sex ratio. Among the surveys, the following data types are missing or not adequately sampled: maturity, fecundity, juvenile life phase, and benthic habitat. Although the programs survey juveniles, no program samples the full extent of juvenile habitats.

b. St. Thomas/St. John

When all five reef fish sampling programs in the St. Thomas/St. John island platform are considered, sampling appears to be island-wide i.e., the geographic coverage is adequate (Table 2c). Combined, the surveys cover depths from 0 – 65 m. SEAMAP-C samples depths up to 100 m for three years in a five-year cycle (Table 2c). Presently, no surveys consistently sample depths greater than 40 m. The planned Reef Fish Survey Expansion pilot study would annually survey to 450 m (Table 2c); however, the funding for this project was rescinded and the start date is to be determined. It is unclear if the sampling frequency is adequate, likely due to logistical concerns, including cancellations during hurricane season and delayed arrival of funds. Among the surveys, the following data types are sampled annually island-wide from 5 – 65 m: relative abundance, proportion occurrence, length composition, and benthic habitat (Table 2c). Age composition and sex ratio are sampled island-wide from 5 – 100 m every 2 – 3 years in a 5-year cycle (Table 2c). Among the surveys, the following data types are missing or not adequately sampled: maturity, fecundity, and juvenile life phase. Although the programs survey juveniles, no program samples the full extent of juvenile habitats.

c. St. Croix

When all four reef fish sampling programs in St. Croix are considered, sampling appears to be island-wide and includes areas of high density, low density, and stratified random sampling; however, continued support is needed on this island platform (Table 2d). Combined, the surveys cover depths of 0 – 65 m. Additionally, SEAMAP-C samples depths up to 100 m for three years in a five-year cycle. The planned Reef Fish Survey Expansion pilot study would annually survey to 450 m (Table 2d); however, the funding for this project was rescinded and the start date is to be determined. Temporal coverage for FI surveys in St. Croix have historically been inadequate. Among the surveys, the following data types are sampled annually island-wide from 0 – 65 m: relative abundance, proportion occurrence, length composition, and benthic habitat. Age composition and sex ratio are sampled island-wide from 5 – 100 m every 2 – 3 years in a 5-year cycle. Among the surveys, the following data types are missing or not adequately sampled: maturity, fecundity, and juvenile life phase. Although the programs survey juveniles, no program samples the full extent of juvenile habitats, specifically the 0 – 1 year old life phase.

d. Overall

Specific needs to ensure adequate data for reef fish are collected for stock assessment include (in no particular order):

- i. Coordination/Calibration of current sampling efforts and standardized survey methods
- ii. Ensure the frequency of sampling is adequate to capture seasonality and periodicity. Input from assessment scientists on frequency of sampling is necessary.
- iii. Continued funding of current programs with developed time series.
- iv. Continued funding of bio-sampling for fishery-independent surveys, some of which is external to “fishery-independent”.

While adult life stages and some juvenile life stages of reef fishes in the US Caribbean are adequately sampled, potential areas for improvement in the current sampling efforts or new sampling efforts may include (in no particular order):

- v. A review of the location and timing of larval fish sampling to identify where additional surveys are needed to properly assess the populations.
- vi. A survey of juvenile habitats e.g. mangroves and shallow seagrass beds (0 – 30 m).
- vii. Establishment of long-term spawning aggregation surveys as most historical projects are short-term.

Beyond the surveys, the development and adoption of standardized maturity assigning protocols to develop size-at-maturity estimates for targeted reef fish species and standardized protocols for fecundity estimation in targeted reef fish species would be useful for stock assessment and potentially help to determine the periodicity of sampling efforts.

## 2. Spiny Lobster

### a. Puerto Rico

There are two main spiny lobster sampling programs in Puerto Rico. One survey has sampled island-wide to depths of 40 meters annually since 2022, and one survey has sampled every five years to depths of 27 m since 2017 (Table 3b). The time series of data for spiny lobster in Puerto Rico is minimal, and the surveys may not cover the entire depth range of the species. As such, it is unclear if the sampling frequency adequately captures population dynamics accurately. From one survey, the following data types are sampled annually island-wide from 0 – 40 m: relative abundance, proportion occurrence, length composition, sex, and maturity (Table 3b). The following data types are missing or not adequately sampled: age composition, fecundity, and benthic habitat. Additionally, pertinent information on adult lobsters has been collected every five years since 2017, while information on juveniles has been collected annually since 2022 (Table 3b).

### b. St. Thomas/St. John and St. Croix

The two spiny lobster surveys in the USVI sample both STTJ and STX, with the same sampling frame (Table 3c, 3d). When both spiny lobster sampling programs in the USVI are considered for both island platforms, one survey has sampled island-wide for one year in a five-year cycle since 2017, while the other has sampled on a 4-year cycle in national parks since 2019 (Table 3c, 3d), i.e., the geographic coverage may not be

adequate. Both surveys sample depths up to 30 meters (Table 3c, 3d). The time series of data for spiny lobster in the USVI is minimal, and the surveys may not cover the entire depth range of the species (Table 3c, 3d). As such, it is unclear if the sampling frequency adequately captures population dynamics accurately. From 0 – 30 m island-wide, the following data types are sampled once in a five-year cycle: relative abundance, proportion occurrence, length composition, age composition, and sex ( Table 3c, 3d). The following data types are missing or not adequately sampled: maturity, fecundity, and benthic habitat. Pertinent data on the target phases of adults and juveniles are limited.

c. Overall

Specific needs to ensure adequate data for spiny lobster are collected for stock assessment include (in no particular order):

- i. Creation of additional surveys to identify potential deeper habitats for lobster and, if found, implementation of surveys to adequately assess the deeper populations that are not inhibited by diving limitations.
- ii. Periodic review of survey frequency to understand if time coverage is adequate, as many programs have only performed pilot studies thus far.
- iii. Incorporation of photographs into current surveys to efficiently estimate fecundity, especially when eggs are present.
- iv. Identification and proper surveillance of juvenile populations and the documentation of lobster virus presence/absence in current surveys to understand population impacts.
- v. Documentation of molt status (in molt, about to molt, or recently molted) island-wide.

3. Queen Conch

a. Puerto Rico

There is one continuous queen conch sampling program in Puerto Rico and one short-term study that began in 2023. Historically, sampling has not been island-wide, and the geographic coverage has not been adequate. The continuous survey has sampled portions of Puerto Rico every five years since 2017, while the other will sample island-wide one time between 2023 – 2025. Combined, the surveys cover 0 – 30 meters of water depth, where queen conch appears to be most abundant. The sampling frequency is inadequate as the time series is minimal. From 0 – 30 m on the east and west coast of the island, the following data types are sampled once in a five-year cycle: relative abundance, proportion occurrence, and benthic habitat. From 0 – 30 m island-wide, length composition data was collected in a short-term pilot study. The following data types are missing or not adequately sampled: length composition, age composition, sex, maturity, and fecundity. Historically, information on 0 – 1 year old queen conch is missing or not adequately sampled.

b. St. Thomas/St. John and St. Croix

The four queen conch surveys in the USVI sample both St. Thomas/St. John and St. Croix with the same sampling frame. When all queen conch sampling programs in the USVI are considered, sampling has occurred island-wide by one survey every five years since 2017 and in national parks every four years since 2019. A short-term study will sample island-wide one time between 2023 – 2025, and another queen conch study

plans to incidentally sample every four years from 2022. Historically, the geographic coverage has not been adequate. Combined, the surveys cover 0 – 30 meters of water depth, where queen conch appears to be most abundant. The sampling frequency may be missing or inadequate. Specifically, seasonality cannot be obtained due to the sampling schedules. From 0 – 30 m island-wide, the following data types are sampled once in a five-year cycle: relative abundance, proportion occurrence, and benthic habitat. From 0 – 30 m and 0 – 25 m in national parks, maturity data are collected once in a 4-year and 3-year cycle, respectively. Among the surveys, the following data types are missing or not sampled island-wide: length composition, age composition, sex, maturity, fecundity, and juvenile life phase. Historically, information on 0 – 1 year old queen conch is missing or not adequately sampled.

c. Overall

Specific needs to ensure adequate data for queen conch are collected for stock assessment include (in no particular order):

- i. Implementation of additional surveys to quantify sex ratios and fecundity
- ii. Identification surveys to quantify the abundance of 0 – 1 year olds, which are typically buried in the sediment
- iii. Implementation of more continuous sampling to increase the frequency and temporal coverage of conch surveys
- iv. Consideration of AI or autonomous vehicles to identify high and low-density areas of interest
- v. Restructured survey approaches and designs such that high and low-density areas are surveyed continuously and repeatedly over time, including consideration of stratified random sampling for low-density areas and targeted surveys in known high-density areas

## C. General Recommendations to Improve US Caribbean Stock Assessment

1. Collaborate with data collection program managers to discuss survey methodology, coverage, overlap, and efficient use of resources to adequately sample target species populations.
  - a. Consider hosting workshops for each survey target with relevant NOAA, SEFSC, and partner staff in attendance
2. Prioritize filling of data gaps based on feasibility given current funding resources and personnel capacity.
  - a. Identify additional funding sources and personnel required to adequately and efficiently fill remaining data gaps
3. Consult with the funding governance CSP group to coordinate additional data collection needs and funding sources.
4. Consult with the toolbox CSP group to understand what additional FI data are required for multi-species or ecosystem-based approaches to fishery management.
5. Coordinate with the US Caribbean stock assessment team to better understand additional gaps regarding the seasonality of sampling as well as target species of interest while taking into consideration constraints of resource allocation and weather conditions.
6. Identify efficient ways to study the role of the species groups in ecosystem health and facilitate Ecosystem Based Fisheries Management.

7. Coordinate with the life history CSP group and stock assessment to ensure surveys are collecting relevant biological samples